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Unit -5 Inventory Control

* Inventory :- An inventory can be defined as a stock of goods which is held for the purpose of future production or sales. The stock of good may be kept in the following forms

(a.) raw-materials (b) partly finished items (c.) finished goods (d) spare parts.

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* Some many Cost :-

JNVU 2012 (1.) The Carrying Cost or Holding Cost or Storage Cost :- The Cost associated with the storage of the inventory until it is sold or used are known as the holding or storage cost. This cost is directly proportional to the increase in the inventory and the time for which the stocks are held. The various components of the holding costs are as follows:

(i) Handling Cost: which includes the cost of labour, transportation charges etc.

(ii) Rent for the space or interest and the cost of depreciation of the space.

(iii) Costs of the staff required to keep records

(iv) Insurance and taxes etc.

(v) Interest on the money locked for inventory

(vi) Deterioration cost etc. which arises in the case of fashion items or items that changes chemically during storage such as medicines, food etc.

It is generally denoted by C_1.

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JNVU 2012

(2.) Shortage Cost or stock-out Cost : It is the Cost that arise due to unfilling of demand in the stock out period. It is usualy of two types
firstly It may be proportional to the quantity short and also to the duration of stock-out ~~It is generally denoted by c_2~~ and includes the cost of lost good will, cost of idle equipement penalty of missing the delivery date etc. It frequently occurs In Cases where back orders are permitted. This is the Case when customers do not Cancel their orders and wait till next lot arrives. Secondly it may be proportional to the quantity short only and is independent of how long stock-out period is. It is most suitable in Case of lost sales where back orders are not permitted and results in a lost sales. This is the Case when customers do not wait and go to some other agency for their requirement. Some times customers withdraw their orders also Causing a major loss due to shortage of one item. It is generally denoted by symbol π.

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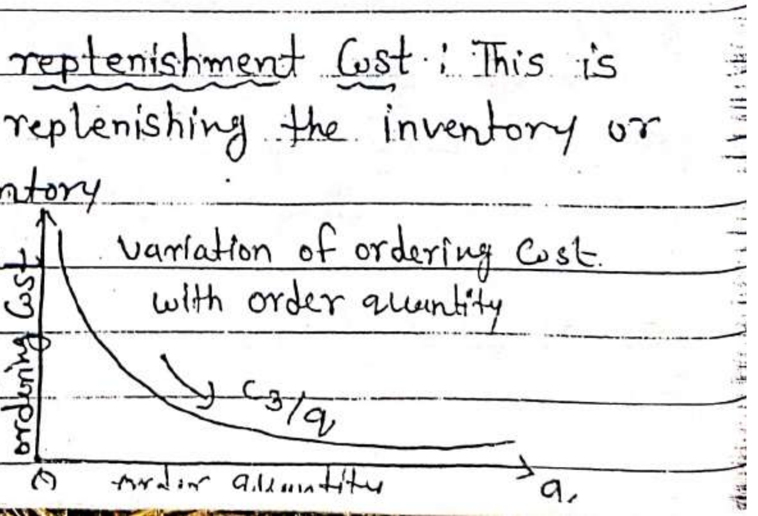
JNVU 2012

(3.) Set-up or ordering or replenishment Cost : This is the Cost incurred in replenishing the inventory or Say changing the inventory
for the ~~exam~~ example the cost of resetting the equipment ~~of~~ for production of fixed administrative



Costs or placing an order etc. It is generally denoted by c_3. and is not proportional to the quantity to be ordered. This cost involves two types of costs, one associated with requisitioning and progressing of order etc. which is same for each set up and every item. on the other hand In case of production policies, the cost of resetting the machine or machines may differ to item. If the quantity ordered is q, then the ordering cost per unit is c_3/q.

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(4.) Procurement Cost :- These are the cost incurred by the inventory system itself in making a procurement of items or in receiving the items to stock. Such costs are directly dependent on the quantity procured. Thus if q units are procured then the cost of procurement may be denoted by $f(q)$ and the average unit cost will be $[f(q)/q]$.

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JNVU 2012

(5.) Purchase price or direct production cost :- It is the actual price at which an item is purchased or procured. In case of production, it is the cost of producing an item. It may be constant or variable both. In many practical situation the unit purchase price depends on the quantity purchased. for instance there may be a discount offer on "big lot size" purchasing. It may reduce due to greater efficiency of men and machines in case of production policy.

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* Some other variables in the Inventory problem.

(1.) Demand :- Demand is the number of items required per period which is not necessarily equal to the amount sold as some demand may go unfulfilled because of shortage or delays.
The demands may be of two types
(i) deterministic demands: If the number of items required in a subsequent period of time is known exactly then such demands are called deterministic demands.
(ii) Non deterministic demands : If the demands over a subsequent period of time is not know with certainly then such demands are called non-deterministic demands.

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(2.) production of goods or supply of goods to the inventory:
The supply of goods to the inventory is also an uncontrolled variable as it may also be deterministic for example supply of water to the pool of dam is deterministic as it depends upon the rainfall in that area, while the supply of items produced by a machine to the inventory is deterministic provided the rate of production is constant.

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(3.) Lead time or Delivery lags or procurement time :-
The time gap between moment of placing an order or deciding for production and the moment

of procurment of items to the inventory against that order is termed as Lead time. It may also be deterministic. It is denoted by Symbol T. If there is no such time gap, then we say that Lead time is zero.

(4.) Types of goods :- The items may be Continuous. Some times items, though Continuous, may be Considered Continuous.

(5.) Buffer stock or Safety Stock -: Generally demand is uncertain and Cannot be pre-determined Completely. Besides this Lead time is also not always exactly known. To overcome the situations of uncertainities in demand and Lead time, Some extra stock is advisable so that shortage may not be there. This extra stock is known as Buffer or safety stock.

(6.) Time horizon -: The time for which an optimal policy is to be determined i.e. over which time cost is to be minimised and inventory level is to be Controlled is termed as time horizon.

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* Some general notations

(i) q = Lot size for one time interval or for one run

(ii) r = Rate of demand = quantity required per unit of time

(iii) k = The rate of production or supply of items to the inventory = rate of replenishment of inventory

(iv) S = A Level of inventory

(v) z = A level of inventory of short items.

(vi) t = Time interval between two Consecutive replenishment of inventory

(vii) R = the total requirement for the time T.

(VIII) $C(q)$ = Total Cost per unit of time as a function of q

(IX) T = Time period in units for which the optimal policy is to be determined (Time horizon).

(X) $p(r)$ = probability density function for r, in case of discrete units of quantity

(XI) $f(r)$ = probability density function for r, in case of continuous unit of quantity.

(XII) q_0, t_0, S_0 = optimal values of q, t, s respectively i.e. the value for which the cost is minimum

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JNVU2008, 12, 13

DETERMINISTIC MODELS

I ⟹ Economic lot size models (single item) OR Harris or wilson Economic lot size formula :-

Sam

State :- Economic lot size with uniform rate of demand :- Consider a period T during which R units of an item are required. The assuptions for this model are as follows;

(1.) Demand is deterministic and is at uniform rate r units of quantity, per unit of time. In such cases we say that the demand is continuous.

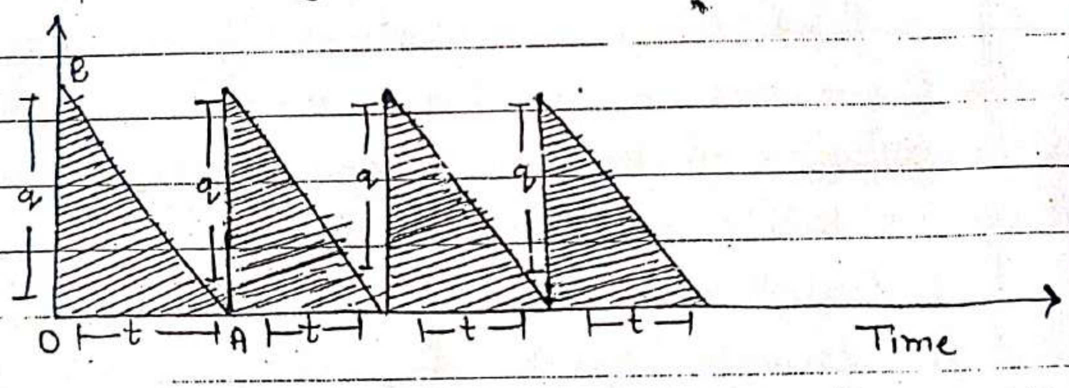
(2.) The Inventory is replenished as soon as the level of the inventory reaches to zero. Thus shortages are not allowed. In such cases we say that shortage cost is infinite so that it cannot be considered in the cost minimisation problem.

(3.) The rate of replenishment of inventory is infinite i.e. the production or supply of items to the inventory is instantaneous or say items are procured in one lot

4. Lead time is zero
5. The lot size is same for each cycle say q, $0<q<\infty$
6. Set up cost is C_3 per cycle
7. Carrying cost is C_1 per unit of quantity per unit of time

Sol. We are to find the optimum value of q which is most economic i.e. gives the minimum total cost.

Let after every time t, the quantity q is ordered throughout the period T, we are to find the optimal values q_0 and t_0 of q and t respectively which minimize the total cost per unit of time. This model involves only C_1 & C_3 costs.



The total inventory in one cycle i.e. for t unit of time

$$= \text{area of the } \Delta OAB = \frac{1}{2} qt$$

$\therefore$ The carrying cost for t units of time $= \frac{1}{2} qtC_1$

Also the set up cost for one cycle i.e. for t units of time $= C_3$

$\therefore$ The total cost in t units of time $= \frac{1}{2} qtC_1 + C_3$

$\therefore$ the total cost for one unit of time

$$= C(q) = \frac{\frac{1}{2} qtC_1 + C_3}{t} = \frac{1}{2} qC_1 + \frac{C_3}{t}$$

But $q = rt$ or $t = \frac{q}{r}$

Therefore $C(q) = \frac{1}{2} qC_1 + \frac{C_3 r}{q}$ is called cost equation

The optimum value of q which minimizes $C(q)$ is obtained by equating the first derivative of $C(q)$ w.r.t. q to zero

Now $\frac{dC(q)}{dq} = \frac{1}{2}C_1 - \frac{C_3 r}{q^2}$

equating it to zero, we get

$$\frac{1}{2}C_1 - \frac{C_3 r}{q^2} = 0$$

$$\boxed{q_0 = q = \sqrt{\frac{2C_3 r}{C_1}}} \quad \text{—(1)}$$

Also $\frac{d^2C(q)}{dq^2} = \frac{2C_3 r}{q^3}$

$$\left(\frac{d^2C(q)}{dq^2}\right)_{q_0} = \frac{2C_3 r}{q_0^3} > 0$$

$\Rightarrow$ $q = q_0$ minimize the cost $C(q)$

eq^n (1) is known as Harris or wilson's economic lot size formula or simply economic lot-size formula and q_0 is also abbreviated as EOQ (Economic order Quantity)

Further $t_0 = \frac{q_0}{r} = \sqrt{\frac{2C_3}{C_1 r}}$

Also minimum cost $= C_0(q_0) = \frac{1}{2}\sqrt{\frac{2C_3 r}{C_1}}\, C_1 + rC_3\sqrt{\frac{C_1}{2C_3 r}}$

$$= \frac{1}{2}\sqrt{(2C_1C_3 r)} + \frac{1}{2}\sqrt{(2C_1C_3 r)}$$

$$C_0(q_0) = \sqrt{(2C_1C_3 r)}$$

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Jnvu 2013

Ex. For an item the production is instantaneous. The storage cost of one time is Re. 1 per month and the set up cost is Rs 25 per ~~month~~ run. If the demand is 200 units per month. find the optimal size of the batch and the best time for the replenishment of inventory.

5 lny

Sol given that

$C_1 = \text{Re } 1.00, \quad C_3 = \text{Rs. } 25, \quad r = 200/\text{months}$

So $q_0 = \sqrt{\frac{2C_3 r}{C_1}} = \sqrt{\frac{2 \times 25 \times 200}{1}} =$

$q_0 = 100$

& $t_0 = \frac{q_0}{r} = \frac{100}{200} = \frac{1}{2}$ month $= 15$ days

So the minimum cost $C(q_0) = \frac{1}{2} q_0 C_1 + \frac{r C_3}{q_0}$

$\frac{r C_3}{q_0}$

$= \frac{1}{2}(100)(1) + 2(25)$

minimum cost $= 50 + 50 = 100$ Rs.

If he produces all the items in the begining of a month then the cost is $= 25 + (1 \times 100)$

$=$ Rs 125

& Average inventory throughout the month $= \frac{q_0}{2} = 100$ items (for month)

Jnvu 2009, 10, 14

II ⇒ Economic lot size with different rates of demand in different cycles. :- In the case all the assumption are same as in the first model except that the demand is uniform with different rates in different cycles.

Sol Suppose $t_1, t_2, t_3 - - - t_n$ are the times of successive cycles & $r_1, r_2, r_3 - - - r_n$ are the demand rates in these cycles ~~repes~~ respectively

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then in the Case we minimize the total Cost for time point T.

The inventory Carrying Cost for the time period T

$\Rightarrow \left(\frac{1}{2} q t_1 + \frac{1}{2} q t_2 + \frac{1}{2} q t_3 + \cdots + \frac{1}{2} q t_n\right) c_1$

$\Rightarrow \frac{1}{2} q c_1 (t_1 + t_2 + \cdots + t_n)$

$\Rightarrow \frac{1}{2} q T c_1$

Since each time q Units are produced, the number of cycles in the total period T is $\frac{R}{q}$, R being the demand of period T

$\{\because R = r_1 + r_2 + r_3 + \cdots + r_n\}$

Therefore the set up Cost for time

$$T = c_3 \left(\frac{R}{q}\right)$$

Hence the total Cost for the time period T is give by $\boxed{C(q) = \frac{1}{2} c_1 q T + c_3 \left(\frac{R}{q}\right)}$ — (1)

the average total Cost per unit time $C(q) = \frac{1}{2} c_1 q + \frac{c_3 R}{Tq}$ — (2)

$\therefore \frac{dC(q)}{dq} = \frac{1}{2} c_1 T - \frac{c_3 R}{q^2}$

for maxima minima $\frac{dC(q)}{dq} = 0$

$\frac{1}{2} c_1 T - c_3 \frac{R}{q^2} = 0$

$\frac{1}{2} c_1 q^2 T - c_3 R = 0$

$$\boxed{q = q_0 = \sqrt{\frac{2c_3 R}{c_1 T}}}$$

which gives the optimal value of q.

$C_0(q_0) = \frac{1}{2} c_1 q_0 + c_3 R / T q_0$ (for unit time)

$= \frac{1}{2} c_1 \sqrt{\frac{2c_3 R}{c_1 T}} + \frac{c_3 R}{T} \times \sqrt{\frac{c_1 T}{2c_3 R}}$

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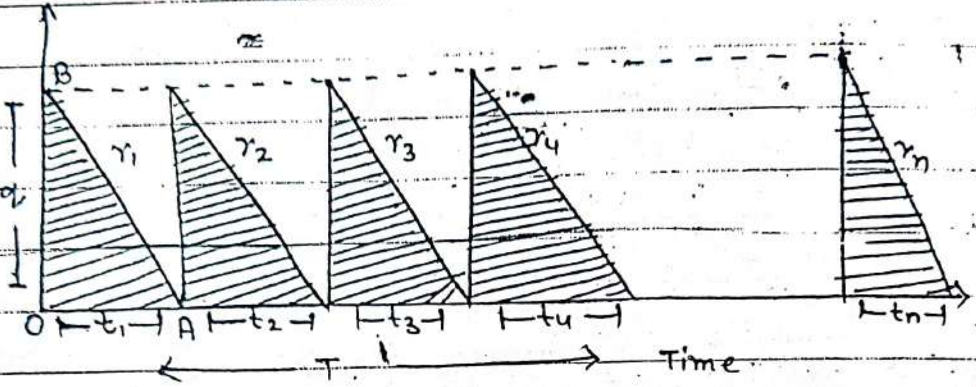
~~$C_0(q_0) = \frac{1}{2}\sqrt{2C_1C_3RT}$~~ ~~$\sqrt{\frac{C_1C_3RT}{2}}$~~



$$C_0(q_0) = \frac{1}{2}\frac{\sqrt{2C_1C_3R}}{\sqrt{T}} + \frac{1}{2}\frac{\sqrt{2C_1C_3R}}{\sqrt{T}}$$

minimum Cost $C_0(q_0) = \sqrt{\left(\frac{2C_1C_3R}{T}\right)}$

It is to be noted that here q_0 is Same as in the first model except that the r, i.e. the Uniform rate of demand is replaced by average rate of demand i.e. by $\frac{R}{T}$.

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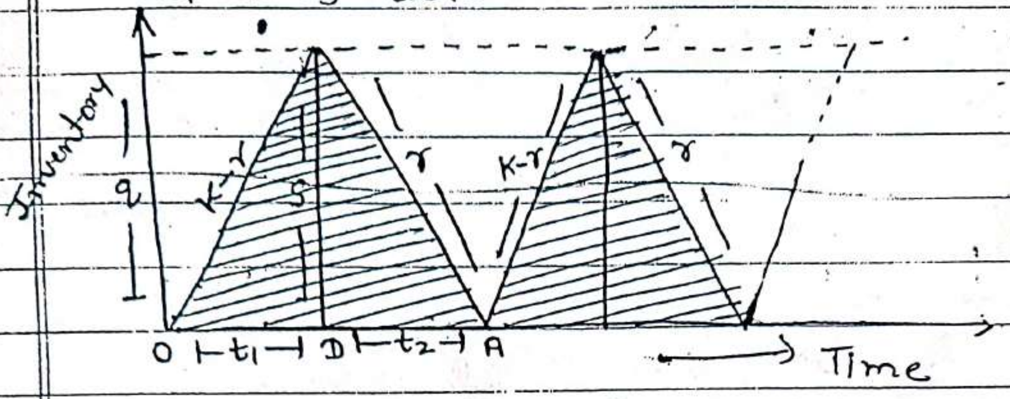
JNVU 2010

III* Economic lot size with finite rate of replenishment or production Lot size model (PLS) :-

Cosider a period T during which R units of an item are required. The assuptions for model are

(1.) ~~...~~ demand is Deterministic and is at uniform rate r units of quantity, per Unit of time
(2.) Shortages are not allowed
(3.) Lead time is zero
(4.) The lot size is Same for each cycle say q.
(5.) Set up Cost is C_3 per cycle & Carrying Cost is C_1 per Unit of quantity per unit of time

Now here we have assumed that replenishment time is zero. This problem also involves only C_1 & C_3 Costs.



It is clear from the fig: that each cycle consists of two parts. In the first part of time i.e. t_1 the inventory is building up (items are assembled) at a constant rate of $(k-r)$ units per unit of time while in the second part t_2 there is only withdrawal of items from the inventory at a constant rate of r units per unit of time. Suppose at the end of the first part of time t_1 the level of inventory is S. Then

$$S = (k-r)t_1 \quad \& \quad S = rt_2$$

$$t_1 = \frac{S}{k-r}, \quad t_2 = \frac{S}{r}$$

Now if q is the lot size then $\boxed{q - t_1 r = S}$

$$\text{or} \quad q - r\frac{S}{k-r} = S$$

$$\Rightarrow \quad q = \left(1 + \frac{r}{k-r}\right)S = \left(\frac{k-r+r}{k-r}\right)$$

$$q = \left(\frac{k}{k-r}\right)S$$

$$\boxed{S = \left(\frac{k-r}{k}\right)q}$$

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Now the Carring Cost for one cycle

$$= \left[\frac{1}{2}(t_1 s) + \frac{1}{2} s t_2\right] c_1$$

$$= \frac{1}{2}(t_1 + t_2) s c_1$$

Hence Carring Cost per unit of time

$$= \frac{\frac{1}{2}(t_1+t_2) s c_1}{t_1+t_2} = \frac{1}{2} s c_1$$

$$= \frac{1}{2}\left(\frac{k-r}{r}\right) q c_1$$

Also the set up cost for one cycle $= c_3$

Set up cost for per unit of time $= \dfrac{c_3}{t_1+t_2}$

$$= \frac{c_3}{\frac{s}{k-r} + \frac{s}{r}} = \frac{c_3 r(k-r)}{s k} = c_3 \frac{r(k-r)}{k} \cdot \frac{k}{(k-r)q}$$

$$= \frac{c_3 r}{q}$$

Therefore total cost of one unit of time

$$C(q) = \frac{1}{2}\left(\frac{k-r}{r}\right) q c_1 + \frac{c_3 r}{q} \quad \text{--- (1)}$$

$$\frac{dC(q)}{dq} = \frac{1}{2}\left(1 - \frac{r}{k}\right) c_1 - \frac{c_3 r}{q^2}$$

Now $\dfrac{dC(q)}{dq} = 0 \Rightarrow \dfrac{1}{2}\left(1-\dfrac{r}{k}\right) c_1 - \dfrac{c_3 r}{q^2} = 0$

(when $C(q)$ is minimum)

$$\boxed{q = q_0 = \sqrt{\frac{2 c_3 r}{c_1(1 - r/k)}}}$$

It gives the required optimum lot size

Also the optimum time for one cycle

$$t_0 = \frac{q}{r}$$

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$$t_0 = \sqrt{\frac{2c_3}{rc_1\left(1-\frac{r}{k}\right)}}$$

by substituting this value of q in ① we find minimum value of cost

$$C_0(q_0) = \frac{1}{2}\left(\frac{k-r}{k}\right)c_1\sqrt{\frac{2c_3 r}{c_1}\frac{1}{(1-r/k)}} + c_3 r\sqrt{\frac{c_1(1-\frac{r}{k})}{2c_3 r}}$$

$$= \frac{1}{2}\sqrt{c_1\left(1-\frac{r}{k}\right)2c_3 r} + \frac{2}{2}\sqrt{\frac{c_1 c_3 r}{2}\left(1-\frac{r}{k}\right)}$$

$$= \frac{1}{2}\sqrt{2c_1\left(1-\frac{r}{k}\right)c_3 r} + \frac{1}{2}\sqrt{2c_1 c_3 r\left(1-\frac{r}{k}\right)}$$

$$\boxed{C_0(q_0) = \sqrt{2c_1 c_3 r\left(1-\frac{r}{k}\right)}}$$

is the minimum cost

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JNVU 2009, 12,

Ex. An item is produced at the rate of 50 items per day. The demand occurs at the rate of 25 items per day. If the set up cost is Rs 100.00 per set up and holding cost is Rs 0.01 per unit of item per day. find the economic lot size for one run. assuming that the shortages are not permitted.

Sol given that

$k = 50/\text{day}$, $r = 25/\text{day}$, $c_1 = \text{Rs } 0.01/\text{day}$

$c_3 = \text{Rs. } 100/\text{run}$

Hence

$$q_0 = \sqrt{\frac{2\times 100\times 25}{0.01\times\left(1-\frac{25}{50}\right)}} = \sqrt{\frac{2\times 100\times \cancel{25}\times 50\times 100}{0.01\times \cancel{25}}}$$

$$q_0 = \sqrt{1000000} = 1000 \text{ items}$$

& $t_0 = \frac{q_0}{r} = \frac{1000}{25} = 40$ days

Now minimum Cost $= \sqrt{\frac{2 \times 0.01 \times 100 \times 25 \times}{100}\left(1 - \frac{25}{50}\right)}$

$= \sqrt{\frac{2 \times 25 \times 25}{50}}$

minimum Cost = Rs 5 per day

~~Hence the total cost for one run~~

$\cancel{C(q_0) = \frac{1}{2}\left(\frac{k-r}{r}\right) q C_1 + \frac{C_3 r}{q}}$

$= \cancel{\frac{1}{2}\left(1 - \frac{25}{50}\right)\frac{1000 \times 0.01}{1000} + \frac{100 \times 25}{1000}}$

$= \cancel{\frac{1}{2} \times \frac{25}{50} \times 10 + \frac{25}{10}}$

Hence total Cost per run = mini. Cost × to

$= 5 \times 40 = \text{Rs } 200.00$

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JNVU 2008, 14

Ex. A Contractor has to supply 10000 paper Cones per day to a textile unit. He find that, when he starts a production run he Can produce 25000 paper Cones per day. The Cost of holding a paper Cone in Stock for one year is 2 paise and the set up cost of a production run is Rs 18. How frequently should production run be made?

Sol Given that $r = 10000$/day

$k = 25000$/day, $C_3 = 18$, $C_1 = \text{Rs } (0.02)$/year

$C_1 = \text{Rs}\left(\frac{0.02}{365}\right)$/day

Now $q_0 = \sqrt{\frac{2 C_3 r}{C_1\left(1 - \frac{r}{k}\right)}}$

$$q_0 = \sqrt{\frac{2 \times 18 \times 10000 \times 365}{0.02 \times \left(1 - \frac{10000}{25000}\right)}}$$

$$= \sqrt{\frac{\cancel{2} \times \cancel{18}^{6} \times 10000 \times 365 \times 25\cancel{000} \times \cancel{100}^{20}}{0.0\cancel{2} \times \cancel{15000}_{5}}}$$

$$= \sqrt{6 \times 10000 \times 365 \times 500}$$

$$= \sqrt{1095000\,0000}$$

$$q_0 = 104642 \text{ (approx)}$$

Now $t_0 = \frac{q_0}{r} = \frac{104642}{10000}$

$t_0 = 10.46$ days.

Hence production of 104642 paper cones should be done after every 10.46 days.

JNVU 2008, 10, 13

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✱ Deterministic models with shortages :-

Model V :- Economic lot-size model with uniform rate of Demand, Infinite rate of production and Having shortages which are to be fulfilled

State :- To derive an economic lot size formula which minimizes the total cost under the following condition

(i) demand is uniform at a rate of r unit per unit time

(ii) production is instantaneous

(iii) Lead time is zero

(iv) c_1 = holding cost per unit per unit time

(v) c_2 = shortage cost per unit per unit time

(vi) c_3 = set up cost per run

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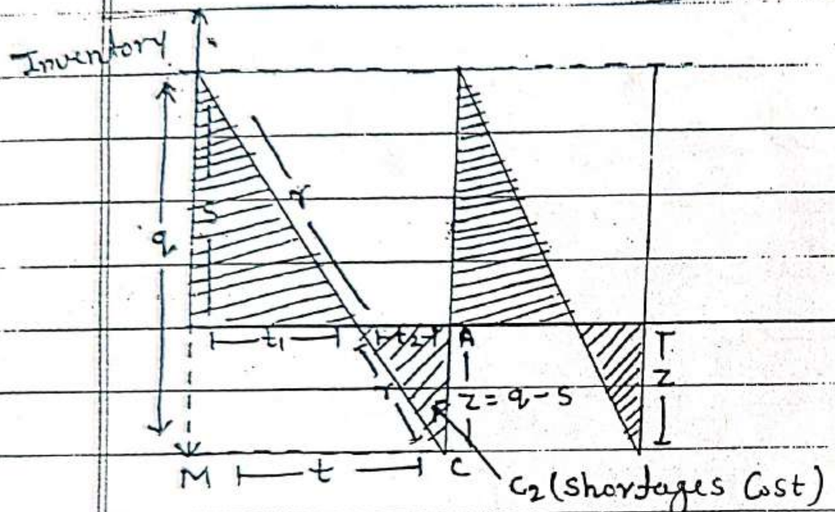
(VII) Shortages are allowed and backlogged.

Sol: Suppose S be the level of inventory in the beginning of each cycle and z is the maximum level of inventory of short items. then

$$q = S + z \Rightarrow \boxed{z = q - S}$$



The inventory Carrying Cost for one cycle

$$= C_1 \times \text{Area of } \Delta OBD = \frac{1}{2} S C_1 t_1$$

Also the Shortage Cost for one cycle

$$= C_2 \times \text{Area of } \Delta DAC = \frac{1}{2} z t_2 C_2$$

$$= \frac{1}{2}(q - S) t_2 C_2$$

Also the set up Cost for one cycle $= C_3$

Hence the total cost for one cycle i.e. for unit of time $= \frac{1}{2} S t_1 + \frac{1}{2}(q-S) t_2 C_2 + C_3$

and the total Cost for per unit of time

$$= \frac{\frac{1}{2} S C_1 t_1 + \frac{1}{2}(q-S) t_2 C_2 + C_3}{t} \quad \text{—(1)}$$

Now obviously $q = rt$ & from the Δ's OBD and Δ's MBC we have $\frac{OD}{OB} = \frac{MC}{MB}$

or $\frac{t_1}{t_2} = \frac{s}{q} \Rightarrow t_1 = \frac{s\,t}{q}$

$$t_1 = \frac{s.t}{rt} = \frac{s}{r} \qquad \{\because q = rt$$

Similarly Δ's DAC and Δ's BCN

$$t_2 = \frac{rt - s}{r}$$

substituting these value of t_1, t_2 and q in (1)

we get

$$c(s,t) = \frac{1}{t}\left[\frac{1}{2} s c_1 . \frac{s}{r} + \frac{1}{2} (rt-s) \frac{(rt-s)}{r}\right] + \frac{c_3}{t}$$

$$= \frac{1}{t}\left[\frac{c_1 s^2}{2r} + \frac{c_2 (rt-s)^2}{2r}\right] + \frac{c_3}{t}$$

$\because$ This is a function of two variables s and t. for its minimum,

$$\frac{\partial c}{\partial s} = 0 \quad \text{and} \quad \frac{\partial c}{\partial t} = 0$$

so $$\frac{\partial c}{\partial s} = \frac{1}{t}\left[\frac{c_1 s}{r} - \frac{c_2 (rt-s)}{r}\right] = 0$$

$$\boxed{s = \frac{c_2 r t}{c_1 + c_2}}$$

Now $$\frac{\partial c}{\partial t} = \frac{-1}{t^2}\left[\frac{c_1 s^2}{2r} + \frac{c_2 (rt-s)^2}{2r} + c_3\right] + \frac{1}{t} c_2 (rt - s) = 0$$

put the value of s in these eqn

$$\Rightarrow -\frac{1}{t^2}\left[\frac{c_1 c_2^2 r^2 t^2}{2r(c_1+c_2)^2} + \frac{c_2}{2r}\left(rt - \frac{c_2 r t}{c_1+c_2}\right)^2 + c_3\right] + \frac{1}{t} c_2 \left(rt - \frac{c_2 r t}{c_1 + c_2}\right) = 0$$

$$\frac{1}{\;}\Big[\, c_1 c_2^2 r t^2 \;\ldots\;\Big] + \ldots$$

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$$\Rightarrow \frac{-c_1 c_2^2 r}{2(c_1+c_2)^2} - \frac{c_1^2 c_2 r}{2(c_1+c_2)^2} - \frac{c_3}{t^2} + \frac{c_1 c_2 r}{(c_1+c_2)} = 0$$

$$\Rightarrow \frac{c_3}{t^2} = \frac{-c_1 c_2^2 r - c_1^2 c_2 r + (c_1+c_2) c_1 c_2 r \times 2}{2(c_1+c_2)^2}$$

$$= \frac{-c_1 c_2^2 r - c_1^2 c_2 r + 2c_1^2 c_2 r + 2c_2^2 c_1 r}{2(c_1+c_2)^2}$$

$$\frac{c_3}{t^2} = \frac{c_1^2 c_2 r + c_2^2 c_1 r}{2(c_1+c_2)^2}$$

$$\frac{c_3}{t^2} = \frac{r c_1 c_2 \cancel{(c_1+c_2)}}{2(c_1+c_2)^{\cancel{2}}}$$

$$t^2 = \frac{2c_3(c_1+c_2)}{c_1 c_2 r}$$

$$\boxed{t = t_0 = \sqrt{\frac{2c_3(c_1+c_2)}{c_1 c_2 r}}}$$

& optimum value of S is

$$S = S_0 = \frac{c_2 r}{c_1+c_2} \times \sqrt{\frac{2c_3(c_1+c_2)}{c_1 c_2 r}}$$

$$S_0 = \sqrt{\frac{2c_2 c_3 r}{c_1(c_1+c_2)}}$$

Therefore the economic lot size is

$$q_0 = r t_0 = r\sqrt{\frac{2c_3(c_1+c_2)}{c_1 c_2 r}}$$

$$q_0 = \sqrt{\frac{2c_3(c_1+c_2)r}{c_1 c_2}}$$

Now $\because z_0 = q_0 - S_0$

$$= \sqrt{\frac{2c_3(c_1+c_2)r}{c_1 c_2}} - \sqrt{\frac{2c_2 c_3 r}{c_1(c_1+c_2)}}$$

$$= \frac{\sqrt{2c_3 r}\,(c_1+c_2) - c_2\sqrt{2c_3 r}}{\sqrt{c_1 c_2(c_1+c_2)}}$$

$$z_0 = c_1\sqrt{2c_3 r}$$

$$\boxed{z_0 = \sqrt{\frac{2C_1C_3r}{C_2(C_1+C_2)}}}$$

—×—×—×—×—×—

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Q. The demand of an item is uniform at a rate of 25 Units per month. The fixed Cost is Rs 15 each time a production is made, The production Cost is ~~RS~~ 1/item, and the inventory Carrying Cost is Rs 0.30/item/month. If the shortage Cost is Rs 1.50/item/month, determine how often to make a production run and of what size it should be?

Sol It is a problem of Deterministic model with shortage. In this problem Set up Cost is a Variable and is equal to $C_3 + Pq$, where P is production Cost of an item and q is the order quantity per production run.

Now Average total Cost per Unit time is given by

$$C(S,t) = \frac{1}{t}\left[\frac{C_1S^2}{2r} + \frac{C_2(rt-S)^2}{2r}\right] + \frac{C_3}{t} + \frac{pq}{t}$$

$$C(S,t) = \frac{1}{t}\left[\frac{C_1S^2}{2r} + \frac{C_2(rt-S)^2}{2r}\right] + \frac{C_3}{t} + pr \quad \left\{\because \frac{q}{t} = r\right\}$$

given that

C_1 = Rs 0.30/item/month, C_2 = Rs 1.50/item/month

C_3 = Rs 15/Set up (fixed)

p = Rs 1.00/item

r = 25/month

$\because$ pr is Constant $\therefore$ as in model, C(S,t) is minimum

Optimal value of q

$$q_0 = \sqrt{\frac{2C_3(C_1+C_2)r}{C_1C_2}} = \sqrt{\frac{2\times15\times(180)\times25\times100}{450\times150\times100}}$$

$q_0 = 10\sqrt{30} = 54$ items.

and optimal time $t_0 = \sqrt{\dfrac{2c_3(c_1+c_2)}{c_1 c_2 r}} = \dfrac{q_0}{r}$

$t_0 = \dfrac{54}{25} = 2.16$ months. Ans

— x — x — x — x — x —

Jnu 2011

* Finite rate of replenishment of inventory :-

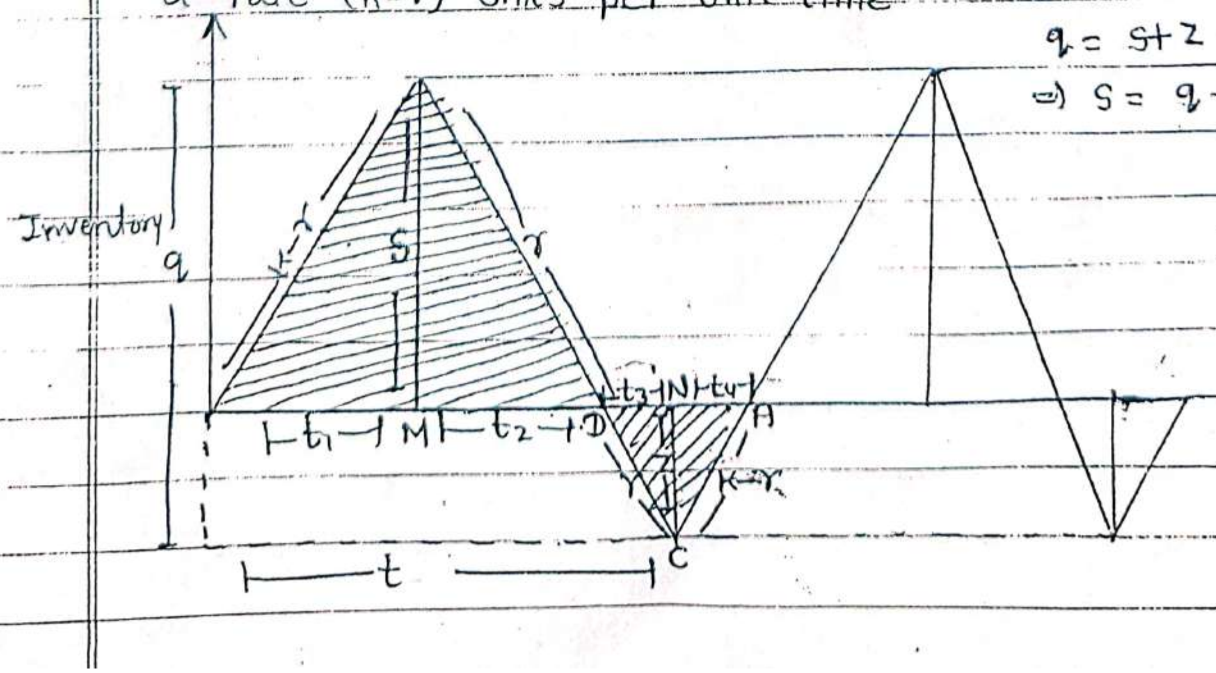
Model VI :- To derive an economic lot-size formula and the average minimum cost per unit time under the following conditions.

(i) demand is uniform at a rate of r units per unit time
(ii) production rate is finite say $k (> r)$ units per unit time
(iii) lead time is zero
(iv) c_1 = holding cost per unit per unit time
(v) c_2 = shortage cost per unit per unit time
(vi) c_3 = set up cost per run, and
(vii) Shortage are allowed and backlogged.

Sol: The stock is zero and the production starts with a finite rate K unit per unit time while the demand is r units per unit time. Thus the inventory increases with a rate $(k-r)$ units per unit time.

$q = S + Z$

$\Rightarrow S = q - Z$



The total cost per cycle for t units of time

$$= \frac{1}{2}(t_1+t_2)sc_1 + \frac{1}{2}(t_3+t_4)zc_2 + c_3$$

But we have

$$s = t_1(k-r) \quad \text{or} \quad t_1 = \frac{s}{k-r}$$

$$s = t_2 r \quad \text{or} \quad t_2 = s/r$$

$$z = t_3 r \quad \text{or} \quad t_3 = z/r$$

$$z = (k-r)t_4 \quad \text{or} \quad t_4 = \frac{z}{k-r}$$

After substituting these value, the cost function is

$$\Rightarrow c_3 + \frac{c_1}{2}\left(\frac{s}{k-r} + \frac{s}{r}\right)s + \frac{c_2}{2}\left(\frac{z}{r} + \frac{z}{k-r}\right)z$$

$$\Rightarrow c_3 + \frac{sc_1}{2}\left(\frac{sr+sk-sr}{r(k-r)}\right) + \frac{zc_2}{2}\left(\frac{zr+zk-zr}{r(k-r)}\right)$$

$$\Rightarrow c_3 + \frac{c_1s^2k + c_2z^2k}{2r(k-r)}$$

$$\Rightarrow c_3 + \frac{c_1s^2 + c_2z^2}{2r(1-r/k)}$$

Therefore the cost per unit of time

$$= c(s,z) \Rightarrow \frac{1}{t}\left[c_3 + \frac{c_1s^2 + c_2z^2}{2r(1-r/k)}\right]$$

If q is the lot size then $q = rt$ or $t = \frac{q}{r}$

$$\therefore c(s,z) = \frac{c_3 r}{q} + \frac{c_1s^2 + c_2z^2}{2q(1-r/k)} \quad \text{--- (1)}$$

Also $s = q - z - r(t_4+t_1)$

$$s = q - z - r\left(\frac{s}{k-r} + \frac{z}{k-r}\right)$$

$$\therefore s\left(1+\frac{r}{k-r}\right) = q - z\left(1+\frac{r}{k-r}\right)$$

$$S = q\left(1 - \frac{r}{K}\right) - z$$

Subtituting this in (1)

$$C(q,z) = \frac{C_3 r}{q} + C_1 \frac{\left[q(1 - r/K) - z\right]^2 + C_2 z^2}{2q(1 - r/K)}$$

$$= \frac{C_3 r}{q} + C_1 \frac{\left[q^2(1-r/K)^2 + z^2 - 2qz(1-r/K)\right] + C_2 z^2}{2q(1-r/K)}$$

Differentiating w.r.t. q & z

Now find $\frac{\partial C(q,z)}{\partial q} = 0$ & $\frac{\partial C}{\partial z} = 0$

$\Rightarrow$ (D.w.r.t q)

$$\frac{\partial C}{\partial q} = -\frac{C_3 r}{q^2} + C_1\left[\frac{2q(1-r/K)^2}{2q(1-r/K)} - \frac{z^2}{2q^2(1-r/K)}\right] - \frac{C_2 z^2}{2q^2(1-r/K)} = 0$$

$$-\frac{C_3 r}{q^2} + \frac{C_1}{2}(1 - r/K) - \frac{C_1 z^2}{2q^2(1-r/K)} - \frac{C_2 z^2}{2q^2(1-r/K)} = 0 \quad \text{--- (2)}$$

$\Rightarrow$ D.w.r to z

$$\frac{\partial C}{\partial z} = C_1\left[\frac{z}{q(1-r/K)} - 1\right] + \frac{C_2 z}{q(1-r/K)} = 0$$

$$C_1 z - C_1 q + \frac{C_1 q r}{K} + C_2 z = 0$$

$$(C_1 + C_2) z = C_1 q(1 - r/K)$$

$$\boxed{z = \frac{C_1 q(1 - r/K)}{(C_1 + C_2)}} \quad \text{in (2)}$$

$$\Rightarrow -\frac{C_3 r}{q^2} + \frac{C_1}{2}\left(1 - \frac{r}{K}\right) - \frac{\cancel{(C_1 + C_2)} C_1^2 \cancel{q^2}(1 - r/K)^{\cancel{2}}}{2\cancel{q^2}\cancel{(1-r/K)}(C_1+C_2)^{\cancel{2}}} = 0$$

$$\Rightarrow \frac{C_3 r}{q^2} = \frac{C_1}{2}\left(1 - \frac{r}{K}\right) - \frac{C_1^2}{2(C_1 + C_2)}\left(1 - r/K\right)$$

$$\frac{c_3 r}{q^2} = \frac{c_1}{2}\left(1-\frac{r}{k}\right)\left(1-\frac{c_1}{c_1+c_2}\right)$$

$$\frac{c_3 r}{q^2} = \frac{c_1}{2}\left(1-\frac{r}{k}\right)\frac{c_2}{(c_1+c_2)}$$

$$q = q_0 = \sqrt{\frac{2c_3(c_1+c_2)r}{c_1 c_2(1-r/k)}}$$

Hence $z_0 = \sqrt{\frac{2c_3 c_1 r(1-r/k)}{c_2(c_1+c_2)}}$

and

$$t_0 = \frac{q_0}{r} = \sqrt{\frac{2c_3(c_1+c_2)}{c_1 c_2 r(1-r/k)}}$$

Now

$$\because S_0 = q_0(1-r/k) - z_0$$

$$S_0 = \left(1-\frac{r}{k}\right)\sqrt{\frac{2c_3(c_1+c_2)r}{c_1c_2(1-r/k)}} - \sqrt{\frac{2c_3c_1r(1-r/k)}{c_2(c_1+c_2)}}$$

$$= \sqrt{\frac{2c_3(c_1+c_2)r(1-r/k)}{c_1c_2}} - \sqrt{\frac{2c_3c_1r(1-r/k)}{c_2(c_1+c_2)}}$$

$$= \frac{\sqrt{c_2+c_1}\sqrt{2c_3(c_1+c_2)r(1-r/k)} - c_1\sqrt{2c_3(1-r/k)r}}{\sqrt{c_1c_2(c_1+c_2)}}$$

$$S_0 = \frac{(c_1+c_2)\sqrt{2c_3r(1-r/k)} - c_1\sqrt{2c_3r(1-r/k)}}{\sqrt{c_1c_2(c_1+c_2)}}$$

$$= \frac{c_2\sqrt{2c_3r(1-r/k)}}{\sqrt{c_1c_2(c_1+c_2)}}$$

$$S_0 = \sqrt{\frac{2c_2c_3r(1-r/k)}{c_1(c_1+c_2)}}$$

Also the minimum value of the cost is

$$C_0(q_0, z_0) = \frac{c_3 r}{q} + \frac{c_1[q(1-r/k)-z]^2 + c_2z^2}{2q(1-r/k)}$$

$$C(q_0, z_0) = \frac{c_3 r \sqrt{c_1 c_2 (1-r/k)}}{\sqrt{2c_3(c_1+c_2) r}} + \frac{c_1\left[\sqrt{\frac{2c_3(c_1+c_2)r}{c_1 c_2} \ldots} - \sqrt{\frac{2c_3 c_1 r (1-r/k)}{c_2(c_1+c_2)}}\right]^2 + c_2\left(\frac{2c_3 c_1 r \ldots}{c_2(c_1 \ldots}\right)}{2q(1-r/k)}$$

$$C(q_0, z_0) = \frac{c_3 r \sqrt{c_1 c_2 (1-r/k)}}{\sqrt{2c_3(c_1+c_2) r}} + \frac{c_1\left[\frac{2c_3(c_1+c_2) r(1-r/k)}{c_1 c_2} + \frac{2c_3 c_1 r(1-r/k)}{c_2(c_1+c_2)} - 2\sqrt{\frac{2c_3(c_1+c_2) r(1-r/k)}{c_1 c_2}} \times \sqrt{\frac{2c_3 c_1 r(1-r/k)}{c_2(c_1+c_2)}}\right] + \frac{2c_2 c_1 c_3 r(1-r/k)}{c_2(c_1+c_2)}}{2q(1-r/k)}$$

$$= \frac{c_3 r \sqrt{c_1 c_2 (1-r/k)}}{\sqrt{2c_3(c_1+c_2) r}} + c_1\left[\frac{2c_3(c_1+c_2) r(1-r/k)}{c_1 c_2} + \frac{2c_3 c_1 r(1-r/k)}{c_2(c_1+c_2)} - \frac{2\times 2 c_3 r(1-r/k)\sqrt{(c_1+c_2)c_1}}{c_2\sqrt{c_1(c_1+c_2)}} + \frac{2c_2 c_1 c_3 r(1-r/k)}{c_2(c_1+c_2)}\right] \Big/ 2q\left(1-\frac{r}{k}\right)$$

$$C(q_0, z_0) = \frac{c_3 r \sqrt{c_1 c_2 (1-r/k)}}{\sqrt{2c_3(c_1+c_2) r}} + \frac{2c_3 r(1-r/k)}{2q(1-r/k)}\left[\frac{(c_1+c_2)\cancel{c_1}}{\cancel{c_1} c_2} + \frac{c_1^2}{c_2(c_1+c_2)} - \frac{2c_1}{c_2} + \frac{\cancel{c_2} c_1}{\cancel{c_2}(c_1+c_2)}\right]$$

$$= \frac{c_3 r \sqrt{c_1 c_2 (1-r/k)}}{\sqrt{2c_3(c_1+c_2) r}} + \frac{c_3 r}{q}\left[\frac{(c_1+c_2)^2 + c_1^2 - 2c_1(c_1+c_2) + c_1 c_2}{c_2(c_1+c_2)}\right]$$

$$= \frac{c_3 r \sqrt{c_1 c_2 (1-r/k)}}{\sqrt{2c_3(c_1+c_2) r}} + \frac{c_3 r}{q}\left[\frac{\cancel{c_1^2} + c_2^2 + 2\cancel{c_1 c_2} + \cancel{c_1^2} - 2\cancel{c_1^2} - 2\cancel{c_1 c_2} + c_1 c_2}{c_2(c_1+c_2)}\right]$$

$$= \frac{c_3 r \sqrt{c_1 c_2 (1-r/k)}}{\sqrt{2c_3(c_1+c_2) r}} + \frac{c_3 r \cancel{c_2}(\cancel{c_2} + c_1)}{q \cancel{c_2}(c_1+\cancel{c_2})}$$

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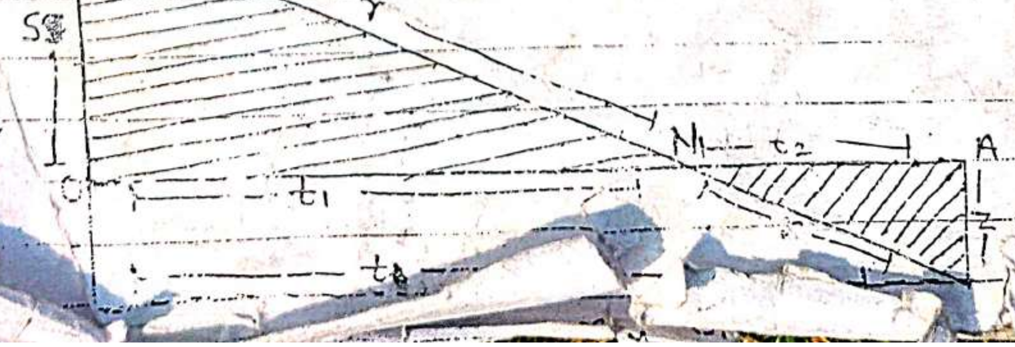
$$\Rightarrow c(q_0, z_0) = c_3 r\sqrt{\frac{c_1 c_2(1-r/k)}{2c_3(c_1+c_2)r}} + \frac{c_3 r(c_2+c_1)}{(c_1+c_2)}\sqrt{\frac{c_1 c_2(1-r/k)}{2c_3(c_1+c_2)r}}$$

$$\Rightarrow c(q_0, z_0) = \frac{c_3 r\sqrt{c_1 c_2(1-r/k)}}{\sqrt{r}\sqrt{2c_3(c_1+c_2)}}\left[1+1\right]$$

$$c(q_0, z_0) = \sqrt{\frac{2 r c_3 c_1 c_2 (1-r/k)}{c_1+c_2}} \quad \text{Ans}$$

---x---x---x---x---

* Fixed time model $\Rightarrow$ Consider a case with instantaneous production and shortages allowed. The inventory is to be replenished after every t time i.e. t is fixed. obviously the model is similar to the model of (Economic lot size with uniform rate of demand). the time period of one cycle is fixed. It is also termed as single period model. The situation of inventory in one cycle is clear.



In this case since t is fixed, $q = rt$ is known exactly.
The decision variable is S. Thus we are to fix up what level the shortage is advisable
obviously the carrying cost for time t is ...
and the shortage cost is $\frac{1}{2}(q-S)\, t_2\, C_2$

But from the Δ's OBN and MBD, we have

$$\frac{t_1}{t} = \frac{S}{q} \Rightarrow t_1 = \frac{S}{q} t$$

Similarly $t_2 = \frac{q-S}{q} t$

The cost for one cycle is

$$C(s) = \frac{1}{2r} C_1 S^2 + \frac{1}{2r} C_2 (q - S)^2$$

$$C(s) = \frac{1}{2r} C_1 S^2 + \frac{1}{2r} C_2 (rt - S)^2 \quad \{\because q = rt\} \quad (2)$$

D. w.r. to S and equating to zero

$$\frac{\partial C}{\partial S} = \frac{C_1 S}{r} + \frac{1}{r} C_2 (rt - S) \times -1 = 0$$

$$C_1 S + C_2 (rt - S) \times -1 = 0$$

$$C_1 S = C_2 rt - C_2 S$$

$$S = \frac{C_2 rt}{C_1 + C_2}$$

Hence $\boxed{S_0 = rt \frac{C_2}{C_1 + C_2}}$

In this model we don't consider the set up cost because it is fixed as the time of one run is fixed.

— x — x — x — x —